



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PASCHIMANCHAL CAMPUS**

**A Minor Project Final Report
On
VIBRATION BASED EARTHQUAKE DETECTION
AND ALERT SYSTEM**

Submitted By:

Kushal Pokhrel (PAS079BEI022)
Mausham Kadariya (PAS079BEI023)
Shashank Singh (PAS079BEI039)
Sujan Raj Regmi (PAS079BEI046)

Submitted To:

Department of Electronics and Computer Engineering
Paschimanchal Campus
Pokhara, Nepal

In partial fulfillment of the requirements for the award of
Bachelor Degree in Electronics, Communication and Information Engineering

Under the Supervision of

Er. Smita Adhikari

December, 2025

DECLARATION

We hereby declare that the project report entitled “Vibration Based Earthquake Detection and Alert System” submitted to the Department of Electronics and Computer Engineering, IOE Paschimanchal Campus, in partial fulfillment of the requirements for the award of the Bachelor Degree in Electronics, Communication and Information Engineering is a record of original work carried out by us under the supervision of our project supervisor. The materials presented in this report have not been submitted elsewhere for the award of any academic degree.

Kushal Pokhrel	(PAS079BEI022)
Mausham Kadariya	(PAS079BEI023)
Shashank Singh	(PAS079BEI039)
Sujan Raj Regmi	(PAS079BEI046)

Date: December, 2025

CERTIFICATE OF APPROVAL

The undersigned certify that they have read and recommended to the Institute of Engineering for acceptance a project report entitled “Vibration Based Earthquake Detection and Alert System” submitted by Kushal Pokhrel, Mausham Kadariya, Shashank Singh, and Sujan Raj Regmi in partial fulfillment of the requirements for the Bachelor’s Degree in Electronics, Communication and Information Engineering.

Supervisor:

Asst. Prof. Smita Adhikari
Department of Electronics and Computer Engineering
IOE, Paschimanchal Campus

External Examiner:

Er. Sangita Lamichhane

Head of Department:

Asst. Prof. Hom Nath Tiwari

Date of Approval: December, 2025

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to the Department of Electronics and Computer Engineering, IOE Paschimanchal Campus, for providing us the opportunity to carry out this minor project.

We extend our heartfelt thanks to our supervisor Er. Smita Adhikari for her continuous guidance, constructive feedback, and valuable suggestions at every stage of the project. Her insights were instrumental in shaping the design and direction of this work.

We also thank all faculty members, our colleagues, and our families for the encouragement and indirect support extended throughout the completion of this project.

Kushal Pokhrel	(PAS079BEI022)
Mausham Kadariya	(PAS079BEI023)
Shashank Singh	(PAS079BEI039)
Sujan Raj Regmi	(PAS079BEI046)

ABSTRACT

This project presents a cost-effective vibration alarm system using two ESP32 microcontrollers and MPU6050 accelerometer sensors. Instead of just one sensor, we used two nodes to cut down on false alarms from nearby disturbances. Node 1 continuously captures acceleration data and smooths it using an exponential moving average (EMA) filter. When the filtered reading exceeds a certain threshold, Node 1 sends a message to Node 2 via ESP-NOW to check if it furthermore noticed the identical vibration. If both nodes agree, the system activates a buzzer and LED, sends an SMS applying the SIM800L module, and revises the Firebase database so users can track the occurrence via a mobile or web dashboard.

We manually applied vibrations under different settings to test the arrangement. Vibration detection, dual-node confirmation, local alarms, SMS delivery, Firebase updates, and cooldown were all successful test scenarios. It is vital to recall that the system can not determine the location, magnitude, or prediction of earthquakes. It functions as an embedded prototype system for vibration warnings at the local level.

Keywords: ESP32, MPU6050, ESP-NOW, GSM, Firebase

1. INTRODUCTION

1.1 Background

Nepal's geography and geological activity make it extremely earthquake-prone. A single seismic occurrence can have a disastrous effect on system, communication networks, and human life, as demonstrated by the devastating Gorkha earthquake in 2015. Numerous buildings fell, communication networks were interfered with, and emergency response processes were harshly delayed throughout the calamity. Numerous priceless assets and houses were moreover lost as a result of the earthquake, highlighting the essential need for dependable initial caution and catastrophe observation systems.

Scientific groups use the professional earthquake tracking system for research reasons. Many people believe that the professional earthquake tracking system is very costly and stays in one center. Because the professional earthquake tracking system requires special rooms and individuals with high education for management, these machines are hard to find. In the land of Nepal, the professional earthquake tracking system is found inside large urban centers while the warning signal is sent by the internet. When electricity and internet lines are cut, the professional earthquake tracking system is stopped. It is known that the warning process fails for people living near or far from the sensors.

A low cost model is built to be placed in every region of the nation so that the warning process is made better. This small machine is utilized to tell neighbors and safety workers about danger. Researchers say that the MEMS sensors provide help for seismic tracking in spots where expensive machines are not located [1]. These small designs are also helpful when detection and warning signals are created [2]. The project is planned for giving warnings in local areas even though the project is not built to take the place of professional tools.

1.2 Problem Statement

The system currently used in Nepal are in the major cities as the deployment is difficult because of their cost and size. A MEMS system can be an alternative at low cost but has inaccuracy and can trigger false alarms.

Using a multi node approach can reduce the false triggering, and the studies [3, 4] have also indicated that this method increases reliability and filtering the noise. This calibration process is done using the installation environment. After all this process, this system activates both, offline and online mode of alert and log those data for further monitoring and research.

1.3 Objectives

To design and implement a cost-effective vibration-based warning system that utilizes a dual-node confirmation strategy.

1.4 Scope of the Project

This project uses a multiple node system using MEMS accelerometers for detecting and confirming vibrations. Both nodes check and verify each other's readings before issuing any alerts. This mechanism is crucial for minimizing false alarms. Valid alerts are communicated through a buzzer, LEDs, SMS, and a mobile application. By utilizing both GSM and Firebase, warnings can still be sent out even if internet access is unavailable.

2. LITERATURE REVIEW

There has been substantial research on using low-cost MEMS accelerometers for earthquake detection and early warning. Chandrakumar et al. provided a systematic review of ground motion sensors, concluding that MEMS accelerometers are practical for early warning systems where professional seismic instruments are not affordable [1]. Abdalzaher et al. explored how cloud computing and IoT communication platforms can facilitate seismic detection and prompt alert delivery [5]. Prasanna et al. presented a node-level detection and local alert generation approach for earthquake early warning, similar to the dual-node structure proposed here [6].

Many studies have looked into distributed sensor networks as a viable alternative to centralized monitoring. Clayton et al. created a Community Seismic Network that demonstrated how dense arrays of low-cost sensors in urban settings can provide valuable seismic data [2]. Faulkner et al. investigated similar community-driven systems and illustrated their capability to detect earthquakes and other rare events through distributed sensor inputs [13]. Picozzi et al. characterized MEMS accelerometers for seismic applications, emphasizing their usefulness while also noting the caution needed when interpreting their outputs [9].

Sensor-level testing and calibration have been examined extensively. Sapkota et al. evaluated a low-cost MEMS accelerometer on a 1D shake table and highlighted the importance of offset calibration and baseline noise analysis before effective ground acceleration data can be captured [7]. Esposito et al. compiled datasets from real seismic events in Central Italy using affordable MEMS instruments [8]. Cascone et al. demonstrated that these sensors can capture high-frequency motion near the source if properly configured [4]. Collectively, these findings indicate that low-cost sensors possess the technological capabilities necessary to serve as the hardware foundation for local alert systems.

Mitigating false alarms through multi-node coordination is an established direction. Nof et al. designed a mini-array using low-cost accelerometers, applying multi-node filtering to eliminate localized noise and enhance trigger accuracy [10]. The MyShake platform extended a similar concept to smartphones, utilizing built-in MEMS sensors from a crowdsourced global network for seismic detection [11].

This project builds upon all these contributions. The integration of dual-node verification, EMA-based noise filtering, GSM backup for connectivity failures, and Firebase for remote monitoring addresses the major gaps identified in the literature for a community-scale low-cost system. Given Nepal's infrastructure context, the GSM fallback is particularly vital, as internet access may not be reliable during seismic events.

3. SYSTEM ARCHITECTURE AND METHODOLOGY

3.1 System Architecture

The system is built with two nodes, each fitted with a microcontroller and an accelerometer. As the primary and principal unit, Node 1 controls an LCD display, sends SMS via the GSM module, uploads data to Firebase, and manages local buzzer and LED alarms. In contrast, Node 2 functions entirely as a validation unit and does not activate warnings on its own.

When Node 1 detects a vibration that exceeds the set threshold, it displays a preliminary alert on the LCD and updates Firebase with the warning status. It then sends a brief message to Node 2 via ESP-NOW, querying whether it registered the same disturbance. If Node 2 confirms it felt the disturbance within 500 ms, the alert gets verified, and the complete alarm sequence is initiated. If not, the event is dismissed as local noise. This two-step verification process helps minimize false alarms.

block_ddiagram.jpg0.85BlockDiagramoftheVBEDASfig : block_ddiagram

3.2 Working Principle

The designed system monitors ground movement constantly without stopping. Here, we used MPU6050 sensor to measure acceleration across 3 different axes: X, Y and Z. The sensed data is then transferred to the processor using I2C protocol. After getting the values of these components, for determining how strong the vibration was, magnitude is calculated using the following equation:

$$A_{mag} = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad (3.1)$$

where A_x , A_y , and A_z represent acceleration along the X, Y, and Z axes respectively.

For removing all the noises and making the system more accurate, the raw data is processed using exponential moving average filter:

$$A_f[n] = \alpha A_r[n] + (1 - \alpha)A_f[n - 1] \quad (3.2)$$

where $A_f[n]$ is the filtered value, $A_r[n]$ is the raw value, and α is the smoothing factor.

When the filtered data passes a predefined threshold, then node 1 first goes in "warning state", then sends a verification signal to node 2. If both nodes detect vibration, then system moves into a "confirmed warning state". By requiring the node verification, we can distinguish between a real emergency and localized disturbance.

3.3 System Flowchart

flowchart.png0.72Operational Flowchart of the Systemfig:flowchart

3.4 Circuit Design and Schematic Diagram

The schematic diagram made using KiCAD shows how all the components/parts are wired together. The ESP32 being the processor, interacting to the accelerometer via I2C and GSM module via UART. To alert people in the limited range from node 1, a buzzer and LED are connected. The design includes a battery and a power switch, allowing the monitoring system to stay active even when not connected to computer.

schematic.png0.95Schematic Diagram of the Vibration Based Earthquake Detection and Alert Systemfig:schematic_{diagram}

3.5 Detection and Verification Logic

To make sure the system works during a real emergency, it follows a two-step detection process. First, Node 1 checks if the vibration detected crossed the predefined threshold or not. If it crossed, then the system sends this signal via wireless communication to Node 2, where this node checks if it also felt that same vibration which is installed at different location inside the range of Node 1. This verification process prevents false alarms caused by the noises affecting only one sensor.

The logical states used in the system are:

1. **NO_VIBRATION**: System is in normal state.
2. **Warning state**: Node 1 detects vibration, displays warning and requests Node 2 for verification.
3. **VIBRATION DETECTED**: Both nodes confirm abnormal vibration and then alert system is triggered giving offline mode priority and simultaneously to online mode.
4. **Cooldown state**: The system then waits for 60 seconds from the time the alert has been triggered.

4. IMPLEMENTATION DETAILS

4.1 Hardware Implementation

The hardware for this system consists of two ESP32, where both of them are connected to MPU6050 sensors to track acceleration data. Here, Node 1 is the primary hub which is responsible for local alerts and remote communication via GSM module and Firebase. The components connected are powered by ESP32, but for GSM module for its operation requires a lot of power to operate, hence a independent battery source is used. This setup ensures the implemented hardware remains stable and has enough power to send alerts without failing.

4.2 PCB Design

For designing the PCB, we used KiCAD. KiCAD was selected because this platform provided user-friendly and offers a faster workflow. The board was meant to be manufactured manually using toner transfer and chemical etching, so kept all the wiring on bottom copper layer for making this process simpler. Solid ground copper plane is also included in the design, making the grounding more efficient but also saves time during chemical process by leaving more copper on the board.

pcb.png0.6PCB Layout Designed in KiCADfig:pcb_{layout}

4.3 Power Supply Implementation

To ensure the system is portable and not requiring a computer, it is powered by rechargeable battery source. A power switch is in series with battery, allowing for easy control over system's power. The ESP32 receives its energy through dedicated voltage source connected to VIN pin, while other components draw power from this component from 3v3 pin (3.3 regulated output voltage). The SIM module on the other hand uses a different power source required for transmitting SMS alerts. This implementation ensures that the rest of the system remains stable.

4.4 Software Implementation

The firmware for this project was developed using Arduino IDE for managing the system's hardware and communication tasks. Upon the connection, program initializes sensors, ESP-NOW used for node to node communication, and setting up connections between GSM and firebase. While the system is active, the ESP32 continuously reads the data from motion sensor, filters it for accuracy and checks with the defined threshold to detect potential vibrations. The two node module are setup and connected with each other for improving system's overall reliability.

4.5 Firebase Integration

The record of all the activity that allows for remote monitoring is kept using Firebase Realtime Database which store event status and sensor readings. The data from the sensor at node 1 is continuously being uploaded to the firebase at a regular time interval. This data serves as a backbone for project's digital interface. This enables both web and mobile applications to access and display real-time safety information to the users.

4.6 GSM Alert Implementation

The SIM module is used for transmitting message signal to the emergency contacts. The ESP32 from node 1 interacts with this module using UART protocol. This module is reliable and can send alert messages without requiring of any internet connection.

4.7 Hardware Prototype

hardware.jpeg0.45Hardware Prototype of the Detection and Alert Systemfig:hardware_{prototype}

The prototype built uses two nodes, each using ESP32 microcontroller and an accelerometer to monitor ground vibrations. The nodes are linked using ESP-NOW protocol, which allows them to verify data with each other instantly. Node 1 is the primary hub The prototype was assembled on breadboards for testing and experimental validation. Jumper wires were used for interconnection between the ESP32 boards, sensors, and communication modules. The hardware arrangement allowed convenient modification and debugging during development and testing. The system was tested under laboratory conditions using simulated vibrations to evaluate sensor response, wireless communication reliability, and alert generation functionality.

5. RESULT AND ANALYSIS

5.1 Testing Conditions

The practical implementation of the prototype was done using manual simulated vibration input. As it was not possible for testing in an industrial setting, so vibration inputs were applied to the sensors. Observations of this system were conducted under normal conditions, single node and two node vibrations, SMS notification, Firebase data storing and updating and local alerting.

5.2 Test Results

Table 5.1: Functional Test Results

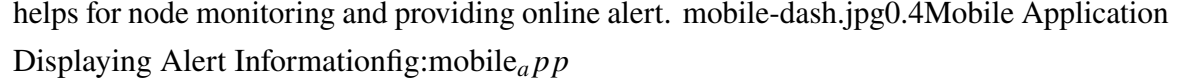
S.N.	Test Case	Expected Output	Result
1	Normal	No alert	Pass
2	Vibration detected by Node 1 only	Check with another node and cancel	Pass
3	Vibration detected signal from both Nodes	Alert confirmed	Pass
4	Inboard (LED and Buzzer) alert activation	Local alert generated	Pass
5	SMS alert	SMS sent to emergency contact	Pass
6	Firebase database update	Event data is updated and stored for monitoring	Pass
7	Cooldown delay after alert	Alert is suppressed after confirmed vibration	Pass

5.3 Firebase Dashboard Output

The Firebase dashboard monitors the node status and event information. When an event is verified by the nodes, the database value is updated and then dashboard status changes from normal to alert state.

firebase.png0.9Firebase Dashboard Displaying Detected Event Datafig:firebase_{dashboard}

5.4 Mobile Application Output

The mobile application, receives the alert information from firebase and displays the alert. It helps for node monitoring and providing online alert.  Mobile Application Displaying Alert Information

5.5 Analysis

From the error analysis table, the system identified vibration when the reading from both the nodes goes above the threshold. The threshold taken while testing on the breadboard was 0.75 but in implementation using PCB, we had to change it to 0.3 as the value predefined required high manual vibration. The values for most of the readings were relatively above the threshold value, it indicates that system can read and identify small as well medium vibrations.

Furthermore, some input were taken near the defined threshold value, such as 0.28112 and 0.32232. The output of these values showed that even minor vibration can trigger alert looking at real world scenarios. In addition, values like 10.57054 and 14.22915, were seen during testing phase because of significant manual input.

In summary, the prototype designed captured all the events that crossed the threshold. However, the reliability depends on appropriate threshold setting (according to the surface), installation of sensors and systems appropriately.

Table 5.2: Measured Acceleration and Threshold Comparison

S.N.	Time	Measured Acc.	Threshold	Difference
1	613 sec	0.4082	0.30	0.1082
2	599 sec	1.0718	0.30	0.7718
3	47 sec	0.3093	0.30	0.0093
4	49 sec	0.6940	0.30	0.3940
5	273 sec	0.4375	0.30	0.1375
6	109 sec	0.6603	0.30	0.3603
7	104 sec	1.4331	0.30	1.1331

6. CONCLUSION

The study presents the design and development of low-cost vibration based earthquake detect and alert system. This prototype was built using ESP32 and MPU6050 accelerometer. The system uses ESP-NOW protocol for node-to-node wireless communication. GSM module is used for SMS alerts and Firebase cloud for monitoring in real time, also sends the data to web and mobile application for online alert. It works on dual node verification approach to avoid false alerts due to noise in the environment. The proposed system operates by monitoring continuously the vibration levels. For maintaining data integrity, raw vibration inputs are smoothed using an exponential moving average filter (low pass filter) and analyzed with the predefined threshold. In case of vibration, the primary node cross checks with the secondary node to verify vibrations. Once a vibration is confirmed from both the nodes, the system deploys a alert response locally through SMS, LED and Buzzer and online through web and mobile application. From the testing phase, we have seen that this embedded system can be used for detecting and providing alerts making it an effective system prototype. However, this model can't be used for predicting earthquake before it occurs, also can't replace the monitoring systems used by the experts.

6.1 System Limitations

The limitations of the designed model system are listed below:

1. The system was tested using manual(hand) simulated vibrations, hence its efficiency during real earthquake events has not been established.
2. The system only detects abnormal vibration and it lacks the specialized instrumentation and algorithms required to estimate magnitude, epicenter and depth of the earthquake.
3. Although the dual node approach minimizes false alarms, severe impacts (such as heavy drilling, construction) within the range may still trigger alerts as earthquake.
4. The alert mechanism depends upon external communication networks. For GSM module to work properly, requires a local GSM cellular towers and for online alert, the ISP infrastructure should be present.

6.2 Future Enhancements

The present model of the earthquake detection and alert system may be enhanced through increased nodes to expand its coverage and make the process more reliable. Using the adaptive threshold algorithm enables the individual node to adjust their sensitivity depending on the particular installation. It will help minimize any noise from the hardware such as structural vibrations or constructions. In the hardware aspect, there could be the inclusion

of a rechargeable battery which will ensure that the system continues working when there is a power outage. Also, solar energy may become one of the improvements to the present design. Moreover, the hardware should have a case to prevent damage from the dust particles. Finally, the GPS modules should be integrated into the system to determine the exact location of each node to facilitate efficient sms notifications and quick response to emergencies. In addition, the system in the user's end may be further developed through upgrading the firebase dashboard to include event history and alert log among others.

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